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Marine animal movement patterns might shape conservation outcomes



Southern sea lion mother and pup on the Falkland Islands.

Animal movement patterns are thought to be dictated by environmental as well as intrinsic factors, such as body length and mass, metabolism, and speed. To determine the influence of intrinsic versus extrinsic factors on animal movement, A. M. M. Sequeira et al. (pp. 3072–3077) analyzed satellite telemetry data on more than 2,600 large marine vertebrates of an array of species with diverse modes of locomotion from 2.8 million locations around the world. In contrast to terrestrial animals, marine vertebrates' effective speed—measured as mean displacement per day across a range of tropical, temperate, and polar habitats—was unaffected by body length or mass, with a few exceptions. Individual variability in horizontal movement patterns was high for species that move along the coasts, which differ in physical features, prey abundance, predatory threats, and human disturbances. By contrast, individual movement variability was low for species that move in the open ocean, which is free of the constraints imposed by continental shelves. Because marine vertebrates such as polar bears, whales, and sharks face anthropogenic threats, uncovering the direct link between habitat complexity and movement patterns might aid conservation efforts in the face of rising sea levels and shrinking Arctic ice. The high variability in behavior of marine animals in coastal habitats also raises the possibility of resilience to deteriorating coastal environments, according to the authors. — P.N.

Explaining the labyrinth-like shape of the nasal cavity

The nasal cavities of animals have similar functions, but they vary widely in shape. Heating and humidifying inhaled air, which are key functions of the nasal cavity, occur more efficiently if the cavity is narrow. However, a narrow nasal cavity has greater resistance to airflow. David Zwicker et al. (pp. 2936–2941) examined how the geometry of the nasal cavity influences airflow and the heating and humidifying of inhaled air. The authors found that the optimal geometry of nasal cavities—with minimal resistance to airflow for a given heating and humidifying efficiency—contains a uniform gap width between the side walls. The overall shape of the nasal cavity has little influence on resistance, and as a result, the cavity can develop labyrinth-like patterns in response to the geometric constraints imposed by the head. The authors found that humans have smaller nasal cavities than expected for their size, and the authors' model explains why humans are forced to breathe through the mouth during heavy exercise. According to the authors, the model suggests that the natural shape



Cross-section of human nasal cavity (drawn to scale).

of the nasal cavity is determined by the trade-off between high exchange efficiency and low resistance to airflow, as well as the geometric constraints imposed by the head. — S.R.

Diversity in tropical reef ecosystems

As coral abundance declines in response to human activity, researchers seek to better predict how decreasing biodiversity will affect tropical reef ecosystems. Although species richness is related to the functional diversity of ecosystems, multiple species can contribute to the same ecosystem functions. Focusing on functional redundancy, Mike McWilliam et al. (pp. 3084–3089) examined how coral species richness affects global patterns of functional diversity based on 821 coral species spanning 12 biogeographical locations across the Indo-Pacific and Atlantic regions. The authors measured seven traits that facilitate key ecosystem functions and constructed a 7D trait space to demonstrate how functional trait diversity changes along gradients of increasing species richness, revealing locations and functions where redundancy is lacking. The authors report that functional diversity remains robust along latitudinal and longitudinal gradients of species richness, declining only in regions with fewer than 200 species and poor redundancy, such as the Persian Gulf, Hawaii, the Caribbean, and the East Pacific. The analysis also highlights the sensitivity of these regions to collapse, as the regions lack key traits or are provided certain traits by a few highly distinctive species with a reduced capacity to adapt to environmental change, according to the authors. — T.J.



A functionally diverse coral community in the Indo-Pacific diversity hotspot. Image courtesy of Tom Bridge (Tethys-images.com).

Vacant land and crime in urban areas

Vacant land comprises about 15% of the land in US cities and has been associated with criminal activity. Urban residents, especially those in low-income neighborhoods, often view vacant land as a threat to personal health and safety. Charles Branas et al. (pp. 2946–2951) examined the relationship between restoration of vacant land plots and crime in Philadelphia. The study included 541 randomly selected vacant lots that were randomly assigned as restoration or control sites. Restoration of the vacant land plots took place in April and May 2013, and the study included crime data from 18 months before and after restoration was completed. Crime data were gathered from police reports and 445 randomly sampled residents living near the land plots. Following restoration of the lots, the authors found that residents who lived near restored land plots had a significantly reduced perception of crime and vandalism as well as an increased feeling of safety and use of outside spaces for relaxing and socializing. Police reports reflected the residents' perceptions and revealed significant reductions in overall crime tied to the interventions. According to the authors, restoration of vacant land might help deter crime and violence in cities. — C.S.

Statistical–experimental approach reveals gene linked to brain atrophy

Genome-wide association studies are a powerful approach to reveal statistical relationships between genetic variants called single-nucleotide polymorphisms (SNPs) and disease-related traits. However, validation and interpretation of the findings remain challenging tasks that are prone to false positives. Marco Lorenzi et al. (pp. 3162–3167) developed a biology-driven validation scheme called functional prioritization and applied the approach to investigate the genetic underpinnings of brain atrophy in Alzheimer's disease (AD). In the statistical discovery phase, the authors identified genetic variants of interest by analyzing relationships between 1.1 million genome-wide SNPs and brain-wide tissue thickness values for 639 healthy individuals and patients using the Alzheimer's Disease Neuroimaging Initiative (ADNI) database. In the first step of the functional prioritization phase, the authors identified a subset of genetic variants involved in relevant biological processes by analyzing gene expression databases. In the second step, the resulting small set of genetic loci shown to modify gene expression was validated in an independent ADNI sample of 553 individuals. Functional prioritization revealed a significant link between the stereotypical pattern of brain atrophy in AD and the *TRIB3* gene, which is known to regulate mechanisms of neurodegeneration. According to the authors, the study illustrates the potential of combining imaging–genetics methods and gene expression data to uncover novel insights into disease-related processes. — J.W.